



SQL Server

Sources

Massachusetts General Hospital Data Warehouse Diagram

Data Warehouse

Consume



CSV Files

- 1- encounters
- 2- patients
- 3- payers
- 4- procedures

Bronze layer



Raw Data

Object Type: Tables
Load:

- Batch Processing
- Full Load
- Truncate & Insert

Transformations:

- None (as-is)

Data Model: None

Silver layer



Cleansed Data

Object Type: Tables
Load:

- Batch Processing
- SCD Type 1 upsert (MERGE) – patients, payers
- Incremental Insert – encounters
- Truncate & Insert – procedures

Transformations:

- Trim & NULL handling / placeholders
- Standardization (Title Case, gender / marital mapping)
- Deduplication (ROW_NUMBER)
- Referential-integrity checks
- Surrogate keys (IDENTITY / lookups)
- IQR outlier capping (durations)
- Derived columns (duration, patient_name)

Data Model: None

Gold layer



Business-Ready Data

Object Type: Views & Tables
Load:

- Batch Processing
- Full Load

Transformations:

- Data Integration
- Business Logic

Data Model: Star Schema (dims + facts)



Power BI Dashboards

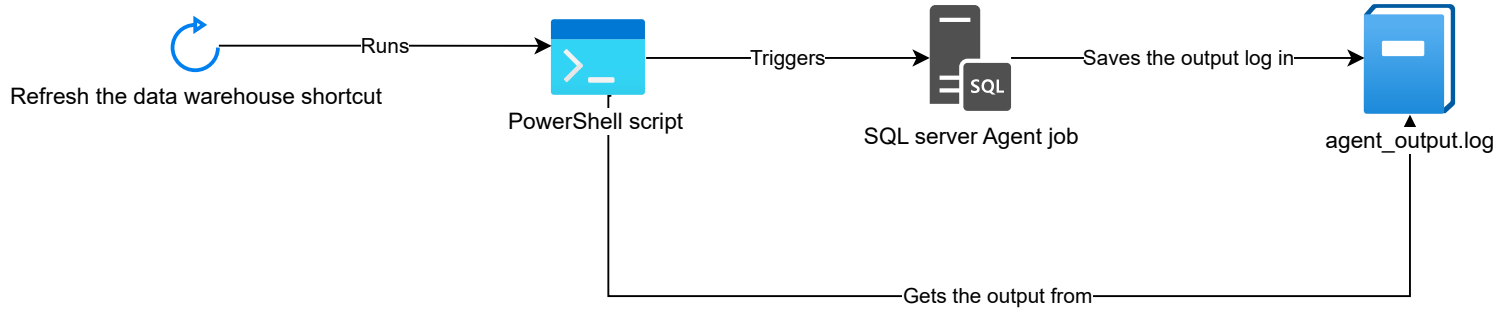


SQL Ad-Hoc Queries

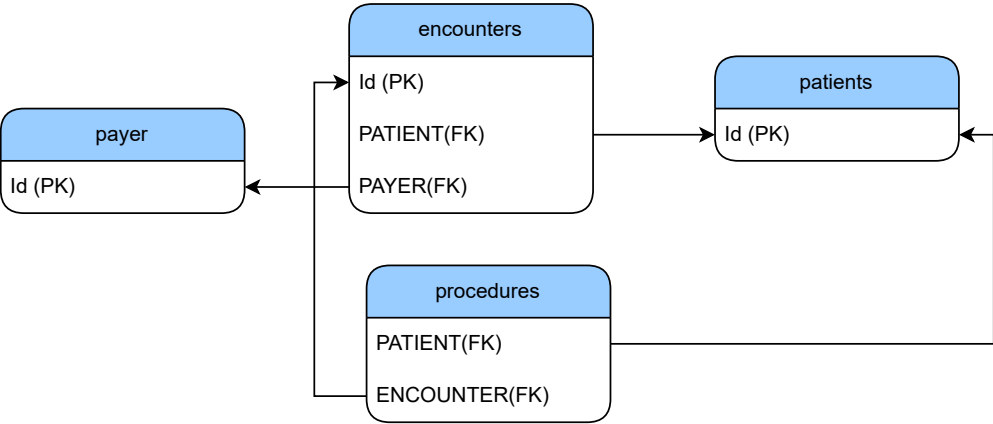


Jupyter Notebooks
(advanced statistical analysis)

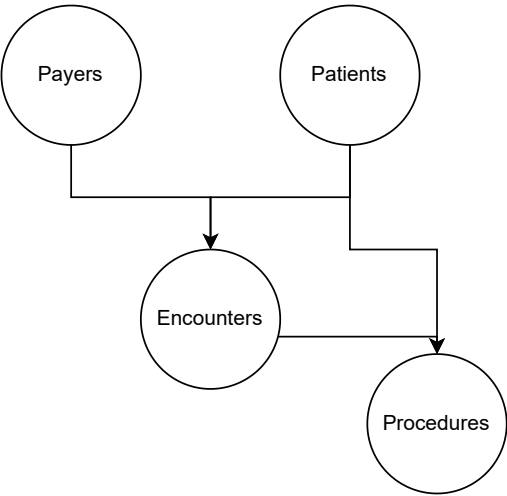
The Data Warehouse Manual Refresh Workflow



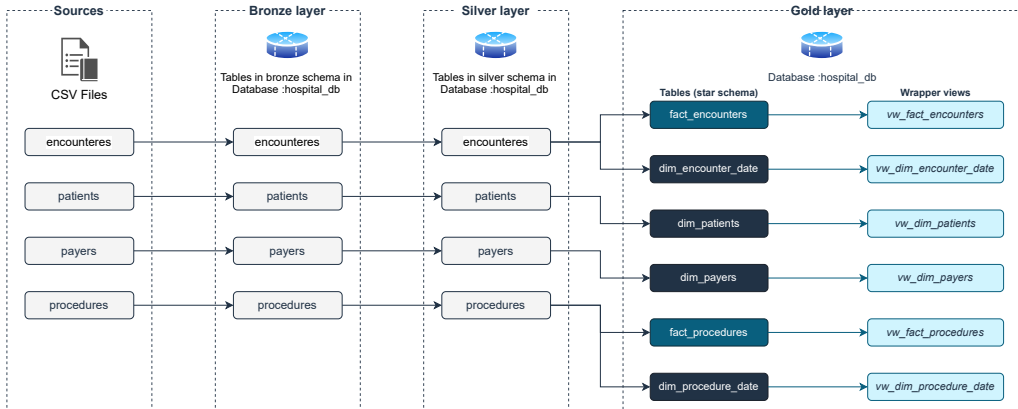
Integration model in the bronz layer



DAG

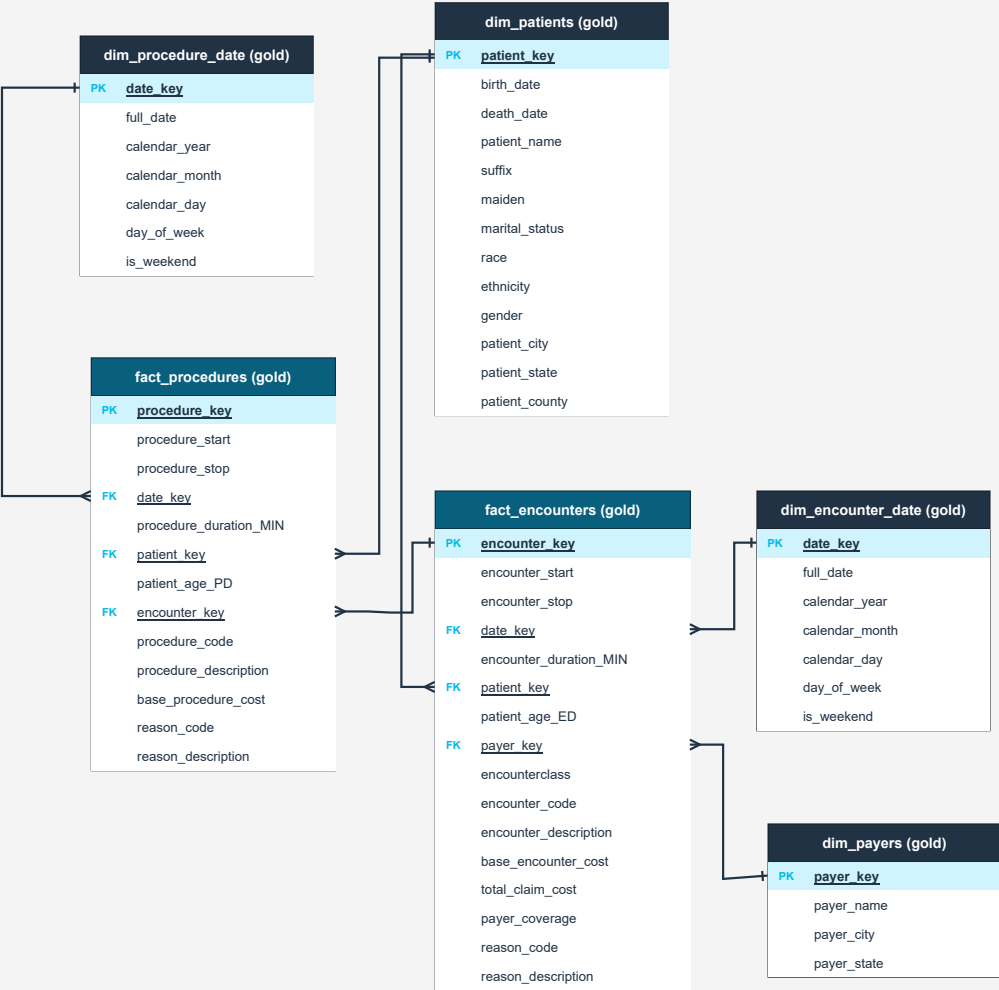


Data Flow Diagram

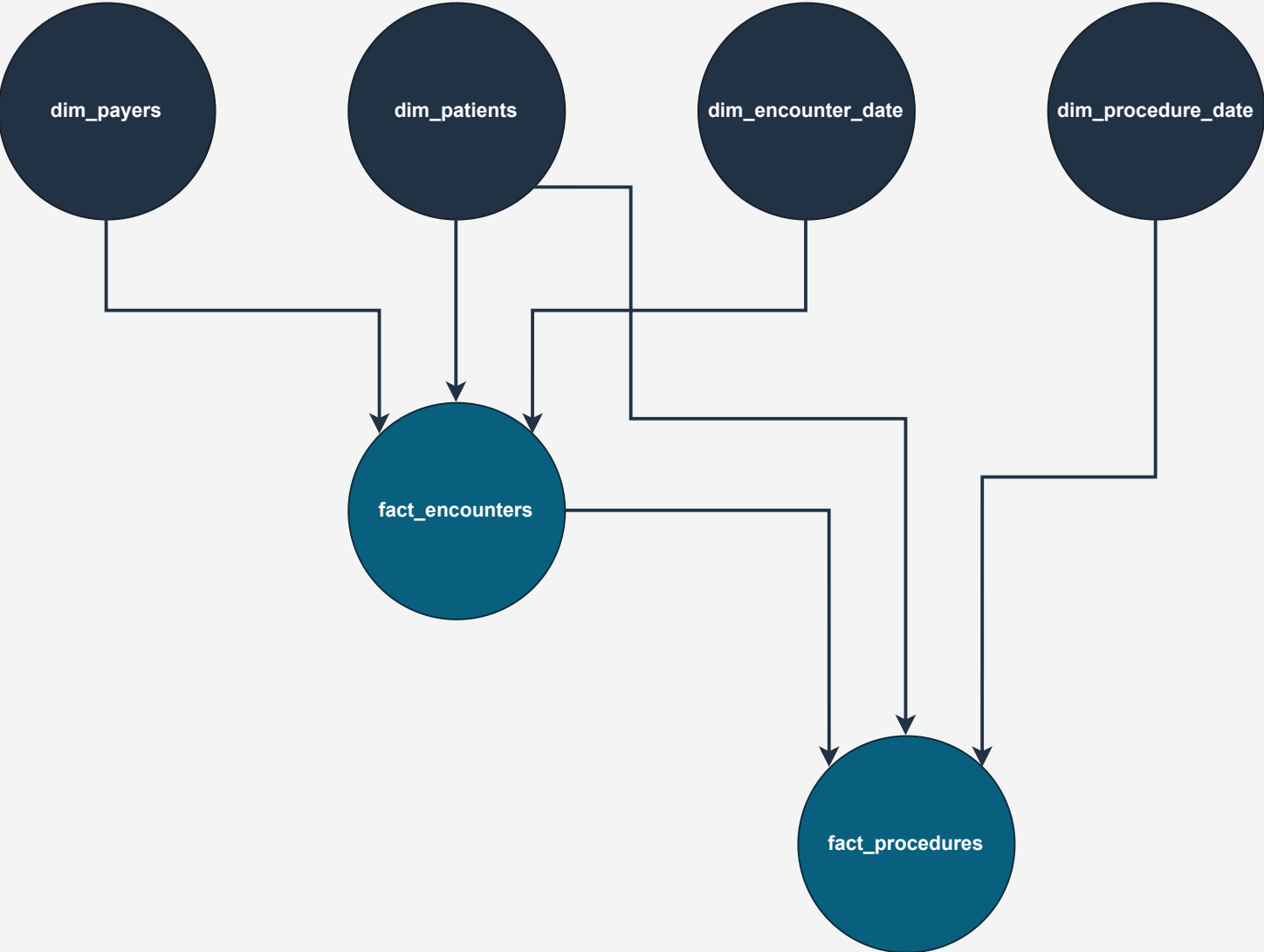


Massachusetts General Hospital — Gold Layer Star Schema

Legend ■ Dimension (navy) ■ Fact (teal) PK = Primary Key FK = Foreign Key



Gold Layer Build DAG (dimensions → facts)



Legend • Dimension (navy) • Fact (teal) arrow: build-order dependency (source built first)

Data Catalog: Gold Layer (Massachussets General Hospital)

Document Overview

- **Layer:** Gold (Business & Analytics Ready)
- **Domain:** Clinical Operations & Payments
- **Refresh Frequency:** Daily (04:00 AM EEST)
- **Data Steward:** Clinical Data Analytics Team

Dimensional Models Reference

dim_patients

- **Description:** Conformed dimension containing mastered patient demographic profiles.
- **Granularity:** One row per unique patient key (`patient_key`).
- **SCD Type:** Type 1 (New data overwrites existing data. No history of the old data).

Schema Definition: gold.dim_patients

Column Name	Data Type	Primary/Foreign Key	Allow Nulls?	Description	Example / Allowed Values
<code>patient_key</code>	INT	PK	No	Unique primary key (Surrogate) for the patient dimension in the Gold layer.	<code>1002948</code>
<code>birth_date</code>	DATE	-	Yes	Patient's date of birth.	<code>1985-06-12</code>